

Engineering Probability

Homework 8

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(1) Joint PMF of Two Discrete Random Variables.

Random variables X and Y have the joint PMF

$$P_{X,Y}(x,y) = \begin{cases} c|x+y|, & x = -2, 0, 2; y = -1, 0, 1, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Find the value of the constant c .

$$\sum_{S_{X,Y}} P_{X,Y}(x,y) = 1$$

$$P(-2, -1) + P(-2, 0) + P(-2, 1) + P(0, -1) + P(0, 0) + P(0, 1) + P(2, -1) + P(2, 0) + P(2, 1) = 1$$

$$c|(-2-1)| + c|(-2+0)| + c|(-2+1)| + c|(0-1)| + c|(0+0)| + c|(0+1)| + c|(2-1)| + c|(2+0)| + c|(2+1)| = 1$$

$$c(|(-3)| + |(-2)| + |(-1)| + |(-1)| + |(0)| + |(1)| + |(1)| + |(2)| + |(3)|) = 1$$

$$c(3 + 2 + 1 + 1 + 0 + 1 + 1 + 2 + 3) = 1$$

$$c(14) = 1, c = \frac{1}{14}$$

(b) Compute $P[Y < X]$.

$$P[Y < X] = P(0, -1) + P(2, -1) + P(2, 0) + P(2, 1)$$

$$P[Y < X] = \frac{1}{14} (|(0-1)| + |(2-1)| + |(2+0)| + |(2+1)|)$$

$$P[Y < X] = \frac{1}{14} (1 + 1 + 2 + 3)$$

$$P[Y < X] = \frac{7}{14}$$

(1) Continued.

(c) Compute $P[Y > X]$.

$$\begin{aligned}P[Y > X] &= P(-2, -1) + P(-2, 0) + P(-2, 1) + P(0, 1) \\P[Y > X] &= \frac{1}{14} (|(-2 - 1)| + |(-2 + 0)| + |(-2 + 1)| + |(0 + 1)|) \\P[Y > X] &= \frac{1}{14} (3 + 2 + 1 + 1) \\P[Y > X] &= \frac{7}{14}\end{aligned}$$

(d) Compute $P[Y = X]$.

$$P[Y = X] = P(0, 0) = \frac{1}{14} (|(0 + 0)|) = 0$$

(e) Compute $P[X < 1]$.

$$\begin{aligned}P[X < 1] &= P[X = -2] + P[X = 0] \\&= P(-2, -1) + P(-2, 0) + P(-2, 1) + P(0, -1) + P(0, 0) + P(0, 1) \\&= \frac{1}{14} (|(-2 - 1)| + |(-2 + 0)| + |(-2 + 1)| + |(0 - 1)| + |(0 + 0)| + |(0 + 1)|) \\&= \frac{1}{14} (3 + 2 + 1 + 1 + 0 + 1) \\&= \frac{8}{14} = \frac{4}{7}\end{aligned}$$

(2) Testing Two Circuits.

Two integrated circuits are tested. In each test, the probability of rejecting the circuit is p . Let X be the number of rejects in the first test and Y the number of rejects in the second test.

- (a) Find the joint PMF $P_{X,Y}(x, y)$.

Possible outcomes for (X,Y) : $(0,0)$, $(0,1)$, $(1,0)$, $(1,1)$

$$P_X(0) = 1 - p, P_X(1) = p$$

$$P_Y(0) = 1 - p, P_Y(1) = p$$

$$P_{X,Y}(x, y) = \begin{cases} p^{x+y}(1-p)^{2-(x+y)} & x = 0, 1; y = 0, 1, \\ 0, & \text{otherwise.} \end{cases}$$

(3) Rejects and Acceptables.

Each test of an integrated circuit produces an acceptable circuit with probability p , independent of other tests. Let N denote the number of circuits rejected in testing r circuits. Let X denote the number of acceptable circuits that appear before the first reject is found.

- (a) Find the joint PMF $P_{N,X}(n, x)$.

$$x = r, n = 0$$

$$P_{N,X}(0, r) = p^r$$

$$P_{N,X}(n, x) = \begin{cases} p^r, & n = 0, x = r, \\ \binom{r-x-1}{n-1} p^{r-n} (1-p)^n, & x = 0, 1 \cdots r-1, n = 1 \cdots r-x \\ 0, & \text{otherwise.} \end{cases}$$

(4) Marginals and Moments.

Given the random variables X and Y in Problem 1:

- (a) Find the marginal PMFs $P_X(x)$ and $P_Y(y)$.

$$P_{X,Y}(x,y) = \begin{array}{c|ccc} x \backslash y & -1 & 0 & 1 \\ \hline -2 & \frac{3}{14} & \frac{2}{14} & \frac{1}{14} \\ 0 & \frac{1}{14} & 0 & \frac{1}{14} \\ 2 & \frac{1}{14} & \frac{2}{14} & \frac{3}{14} \end{array}$$

$$P_X(x) = \begin{cases} \frac{6}{14}, & x = -2 \\ \frac{2}{14}, & x = 0 \\ \frac{6}{14}, & x = 2 \\ 0, & \text{otherwise.} \end{cases} \quad P_Y(y) = \begin{cases} \frac{5}{14}, & y = -1 \\ \frac{4}{14}, & y = 0 \\ \frac{5}{14}, & y = 1 \\ 0, & \text{otherwise.} \end{cases}$$

- (b) Compute the expected values $E[X]$ and $E[Y]$.

$$E[X] = (-2) \left(\frac{6}{14} \right) + (0) \left(\frac{2}{14} \right) + (2) \left(\frac{6}{14} \right) = 0$$

$$E[Y] = (-1) \left(\frac{5}{14} \right) + (0) \left(\frac{4}{14} \right) + (1) \left(\frac{5}{14} \right) = 0$$

- (c) Compute the standard deviations σ_X and σ_Y .

$$\sigma_X = \sqrt{\text{Var}[X]} = \sqrt{E[X^2] - (E[X])^2} \quad \sigma_Y = \sqrt{\text{Var}[Y]} = \sqrt{E[Y^2] - (E[Y])^2}$$

$$E[X^2] = (-2)^2 \left(\frac{6}{14} \right) + (0)^2 \left(\frac{2}{14} \right) + (2)^2 \left(\frac{6}{14} \right) = \frac{48}{14} = \frac{24}{7}$$

$$E[Y^2] = (-1)^2 \left(\frac{5}{14} \right) + (0)^2 \left(\frac{4}{14} \right) + (1)^2 \left(\frac{5}{14} \right) = \frac{10}{14} = \frac{5}{7}$$

$$\sigma_X = \sqrt{\left(\frac{24}{7} \right) - (0)^2} \quad \sigma_Y = \sqrt{\left(\frac{5}{7} \right) - (0)^2}$$

$$\sigma_X = \sqrt{\frac{24}{7}} \quad \sigma_Y = \sqrt{\frac{5}{7}}$$

$$\sigma_X \approx 1.852 \quad \sigma_Y \approx 0.845$$

(5) Function of Two Variables.

For the random variables X and Y in Problem 1, define $W = X + 2Y$.

- (a) Find the probability mass function $P_W(w)$.

Possible values of w :

x	y	$P(x, y)$	w
-2	-1	$\frac{3}{14}$	-4
-2	0	$\frac{2}{14}$	-2
-2	1	$\frac{1}{14}$	0
0	-1	$\frac{1}{14}$	-2
0	0	$\frac{0}{14}$	0
0	1	$\frac{1}{14}$	2
2	-1	$\frac{1}{14}$	0
2	0	$\frac{2}{14}$	2
2	1	$\frac{3}{14}$	4

$$P_W(w) = \begin{cases} \frac{3}{14}, & x = -4 \\ \frac{3}{14}, & x = -2 \\ \frac{2}{14}, & x = 0 \\ \frac{3}{14}, & x = 2 \\ \frac{3}{14}, & x = 4 \\ 0, & \text{otherwise.} \end{cases}$$

- (b) Compute $E[W]$.

$$E[W] = (-4) \left(\frac{3}{14} \right) + (-2) \left(\frac{3}{14} \right) + (0) \left(\frac{2}{14} \right) + (2) \left(\frac{3}{14} \right) + (4) \left(\frac{3}{14} \right) = 0$$

- (c) Compute $P[W > 0]$.

$$P[W > 0] = P_W(2) + P_W(4)$$

$$P[W > 0] = \frac{3}{14} + \frac{3}{14}$$

$$P[W > 0] = \frac{6}{14}$$

$$P[W > 0] = \frac{3}{7}$$

(6) Joint Dependence and Correlation.

For the random variables X and Y in Problem 1:

$$P_{X,Y}(x, y) = \begin{cases} \frac{1}{14} |x + y|, & x = -2, 0, 2; y = -1, 0, 1, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Compute the expected value of $W = 2XY$.

x	y	$P(x, y)$	w	$P_W(w) = \begin{cases} \frac{6}{14}, & w = -4 \\ \frac{6}{14}, & w = 0 \\ \frac{2}{14}, & w = 4 \\ 0, & \text{otherwise.} \end{cases}$
-2	-1	$\frac{3}{14}$	4	
-2	0	$\frac{2}{14}$	0	
-2	1	$\frac{1}{14}$	-4	
0	-1	$\frac{1}{14}$	0	
0	0	$\frac{0}{14}$	0	
0	1	$\frac{1}{14}$	0	
2	-1	$\frac{1}{14}$	-4	
2	0	$\frac{2}{14}$	0	
2	1	$\frac{3}{14}$	4	

$$E[W] = \sum_{S_W} w P_W(w) = (-4) \frac{6}{14} + (0) \frac{6}{14} + (4) \frac{2}{14} = \frac{-16}{14} = \frac{-8}{7}$$

(b) Compute the correlation $E[XY]$.

$$E[XY] = \frac{1}{2} E[2XY] = \frac{1}{2} \left(\frac{-8}{7} \right) = \frac{-4}{7}$$

(c) Compute the covariance $\text{Cov}[X, Y]$.

$$\text{Cov}[X, Y] = E[XY] - E[X]E[Y] = \frac{-4}{7} - (0)(0) = \frac{-4}{7}$$

(d) Compute the correlation coefficient $\rho_{X,Y}$.

$$\rho_{X,Y} = \frac{\text{Cov}[X, Y]}{\sigma_X \sigma_Y} = \frac{\frac{-4}{7}}{\left(\sqrt{\frac{24}{7}} \right) \left(\sqrt{\frac{5}{7}} \right)} = \frac{-4}{\sqrt{120}} \approx -0.365$$

(6) Continued.

(e) Compute the variance $\text{Var}[|X + Y|]$.

x	y	$P(x, y)$	z	$P_Z(z) = \begin{cases} \frac{4}{14}, & z = 1 \\ \frac{4}{14}, & z = 2 \\ \frac{6}{14}, & z = 3 \\ 0, & \text{otherwise.} \end{cases}$
-2	-1	$\frac{3}{14}$	3	
-2	0	$\frac{2}{14}$	2	
-2	1	$\frac{1}{14}$	1	
0	-1	$\frac{1}{14}$	1	
0	0	$\frac{0}{14}$	0	
0	1	$\frac{1}{14}$	1	
2	-1	$\frac{1}{14}$	1	
2	0	$\frac{2}{14}$	2	
2	1	$\frac{3}{14}$	3	

$$E[Z] = E[|X + Y|] = 1 \cdot \frac{4}{14} + 2 \cdot \frac{4}{14} + 3 \cdot \frac{6}{14} = \frac{30}{14}$$

$$E[Z^2] = E[|X + Y|^2] = 1^2 \cdot \frac{4}{14} + 2^2 \cdot \frac{4}{14} + 3^2 \cdot \frac{6}{14} = \frac{74}{14}$$

$$\text{Var}[|X + Y|] = E[|X + Y|^2] - (E[|X + Y|])^2 = \frac{74}{14} - \left(\frac{30}{14}\right)^2 = \frac{34}{49} \approx 0.694$$

(7) Dependent Support Random Variables.

Random variables X and Y have the joint PMF

$$P_{X,Y}(x, y) = \begin{cases} \frac{1}{21}, & x = 0, 1, 2, 3, 4, 5; \ y = 0, 1, \dots, x, \\ 0, & \text{otherwise.} \end{cases}$$

1. Find the marginal PMFs $P_X(x)$ and $P_Y(y)$.

all possible values of (x,y): Are equally likely. Simply a matter of counting number of x=0s, x=1s etc and same for y. x can take any number of values of y up to x so its simply x plus the value itself or x+1. y can take any number of values of x so it is total number of x values (6) minus y.

$$P_X(x) = \begin{cases} \frac{x+1}{21}, & x = 0, 1, 2, 3, 4, 5; \\ 0, & \text{otherwise.} \end{cases} \quad P_Y(y) = \begin{cases} \frac{6-y}{21}, & y = 0, 1, 2, 3, 4, 5; \\ 0, & \text{otherwise.} \end{cases}$$

2. Find the covariance $\text{Cov}[X, Y]$.

$$E[XY] = \sum_{S_{X,Y}} xy P_{X,Y}(x, y)$$

$$E[XY] = \sum_0^5 \sum_0^x xy \left(\frac{1}{21} \right) = \frac{1}{21} \sum_0^5 x \left(\frac{x(x+1)}{2} \right) = \frac{1}{42} \sum_0^5 x^2 + x^3$$

$$E[XY] = \frac{1}{42} ((0)^2 + (0)^3 + (1)^2 + (1)^3 + (2)^2 + (2)^3 + (3)^2 + (3)^3 + (4)^2 + (4)^3 + (5)^2 + (5)^3)$$

$$E[XY] = \frac{1}{42} (1 + 1 + 4 + 8 + 9 + 27 + 16 + 64 + 25 + 125) = \frac{280}{42}$$

$$E[X] = \sum_{S_X} x P_X(x)$$

$$E[Y] = \sum_{S_Y} y P_Y(y)$$

$$E[X] = \sum_{x=0}^5 x \left(\frac{x+1}{21} \right) = \frac{1}{21} \sum_{x=0}^5 (x^2 + x)$$

$$E[Y] = \sum_{y=0}^5 y \left(\frac{6-y}{21} \right) = \frac{1}{21} \sum_{y=0}^5 (6y - y^2)$$

$$E[X] = \frac{70}{21} = \frac{10}{3}$$

$$E[Y] = \frac{35}{21} = \frac{5}{3}$$

$$\text{Cov}[X, Y] = \left(\frac{280}{42} \right) - \left(\frac{10}{3} \right) \left(\frac{5}{3} \right) = \frac{10}{9} \approx 1.11$$